Tanmay Bhuskute

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Cincinnati, OH

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience designing and operating distributed systems at scale in Java and Python, including deployment platforms, orchestration workflows, and high-availability infrastructure on AWS and Kubernetes.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, Go, JavaScript, TypeScript, SQL.

Distributed Systems: Microservices Architecture, REST APIs, System Design, Design Patterns, High Availability.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS (S3, EC2, Lambda), Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, SonarQube.

Testing & Quality: Unit Testing, Integration Testing, Code Review, Test-Driven Development, JUnit, Pytest.

WORK EXPERIENCE

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** distributed infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live large-scale infrastructure issues across Linux, EC2, databases, and Teamcenter X services with full ownership on-call.
- Designed **Ansible-based Blue-Green deployment** framework improving system reliability, reducing production upgrade downtime by **50%** with automated rollback.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault to modernize architecture at scale.
- Served as Quality Coach and SonarQube Specialist, enforcing code review standards and raising automated test coverage from **35% to 78%**; mentored **5 software engineers**.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production through automated testing.

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** baseline across 53 runs on 20+ apps, demonstrating **25.7x latency reduction** while preserving task-completion fidelity.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built large-scale ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash with Zod schema validation at every boundary, eliminating a class of runtime type-mismatch bugs across all external tool calls.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).